

complex motions and animations can be achieved. Much like with modifiers on the modifier stack, controllers can also be copied and pasted to copy animation. Controllers can also be instanced so that changes in the animation of one object affect another.

Controllers are automatically added to a parameter when it is key framed. These parameters are typically the parameters involved with transformations (position, rotate, scale) but can be any animatable parameter. Controllers can also be added to existing objects to modify the way the objects are animated. Controllers are found under the Animation option of the main toolbar. They can also be assigned and managed in the Curve Editor, the Dope Sheet, and the Motion panel.

7.2.1 Keyframe controllers

The most common type of controller manages the key framing of an object's parameters. These keyframes show up on the time slider as well as in the Curve Editor and Dope Sheet, where they can be edited. The Bezier float controller is the most common of the controllers. It allows motion to be keyframed with the in-betweens defined by a Bezier curve. Linear interpolate the animation linearly, creating a straight line in between keyframes. TCB controllers produce curve-based animation, but do not use Bezier tangents or adjustable tangent handles. These controllers use fields to adjust the Tension, Continuity, and Bias of the animation. The TCB controller creates smoother rotations but does not allow for curve editing.

7.2.2 Parametric controllers

Some controllers use a set of user-defined parameters to create motion. These controllers do not need key frames to operate; after the parameters are set, the animation proceeds on its own.

The Audio controller converts the amplitude of a sound file or real-time sound wave to values that can drive the animation of almost any parameter. This is great for syncing animation to sound. The Noise controller uses random noise to create animation. The strength and frequency of the noise can be adjusted. The spring controller adds spring like dynamic effects to any point or object position, to create secondary motion. This constraint adds realism to generally static animations.

The Waveform controller creates periodic waveforms. This is good for any type of regular animation, such as a blinking light or a pendulum.